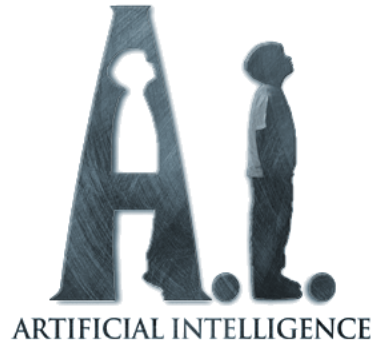


COSC2129/1476

Artificial Intelligence

The End
Can machines think?
AI @ RMIT





Let's Recap



ARTIFICIAL INTELLIGENCE



Did I tell you these was going to be like...



Why AI is an important course?

1. It provides the **core knowledge** of computer science.
 - Hence, serious programmers and software engineers should know about major AI techniques.
2. It can be **profitable**.
3. It is also **fun**.
4. It relates to other **non-exact sciences**.
5. It is **different** to most other subjects.

What have we seen in the course?

Course objectives: `Basic concepts and techniques of AI`

1. **Search:** as a general problem technique.
2. **Knowledge Representation (KR&R):** rational behavior requires knowledge! Beyond databases...
3. **Automated Planning:** what plan should I execute? Search + KR
4. **MDP:** decision making under uncertainty
5. **Reinforcement Learning:** learn environment and how to act rationally.
6. **Probabilities:** basic tool for reasoning under uncertainty.
7. **Bayesian Networks:** knowledge representation and reasoning for probabilistic reasoning.

So, what is “AI”?

“Artificial intelligence is that activity devoted to making machines intelligent, and intelligence is that quality that enables an entity to function appropriately and with foresight in its environment.” — Nils J. Nilsson



Defining AI

- Preface
- Executive Summary
- Overview
- Section I: What is Artificial Intelligence?**
 - Defining AI**
 - The Human Measure
 - An Operational Definition
 - AI Research Trends
- Section II: AI by Domain



AI00 | September 28, 2021 |

Curiously, the lack of a precise, universally accepted definition of AI probably has helped the field to grow, blossom, and advance at an ever-accelerating pace. Practitioners, researchers, and developers of AI are instead guided by a rough sense of direction and an imperative to “get on with it.” Still, a definition remains important and Nils J. Nilsson has provided a useful one:

“Artificial intelligence is that activity devoted to making machines intelligent, and intelligence is that quality that enables an entity to function appropriately and with foresight in its environment.”^[3]



You say more?

How to achieve “intelligent” behavior through computational means.

Thus we try to construct systems whose **computation** achieves or approximates the desired notion of **rationality**:

Rationality: Precise mathematical notion of **what it means to do the right thing** in any particular circumstance.

Four AI Definitions by R&N

	Like Humans	Not necessary like humans
Think	Systems that think like humans	Systems that think rationally
Act	Systems that act like humans	Systems that act rationally



Cognitive Science



OUR FOCUS

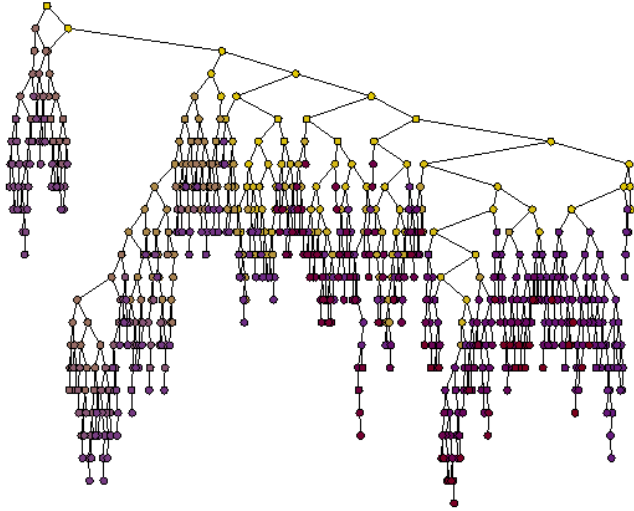


Our Artificial Intelligence Course

Course objectives: `Basic concepts and techniques of AI`

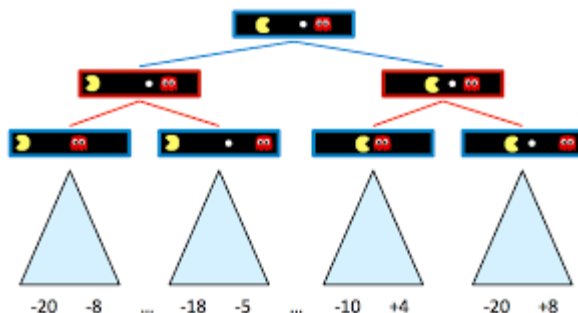
- Search.
- Knowledge representation.
- Planning.
- Intelligent Agents.
- Reasoning under uncertainty.
- Machine Learning.

Search

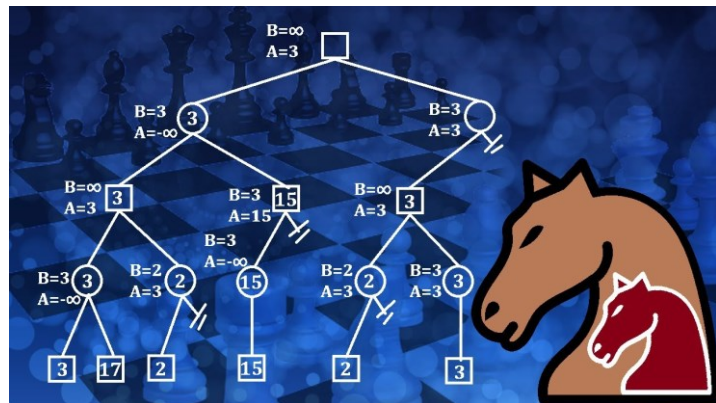


- Search is an exploration of possibilities.
- General problem solving tool: useful when a 'direct' method/algorithm is not known.
- Central component to many AI systems.
- Mathematically: "Graph reachability"
 - Possible states
 - Initial state
 - Actions
 - Transition model
 - Goal test/condition
- Algorithms: DFS, BFS, Best FS, IDS, A*, Greedy Best FS, etc..
 - Algo properties & complexity
- Heuristics to guide the search.

Adversarial Search

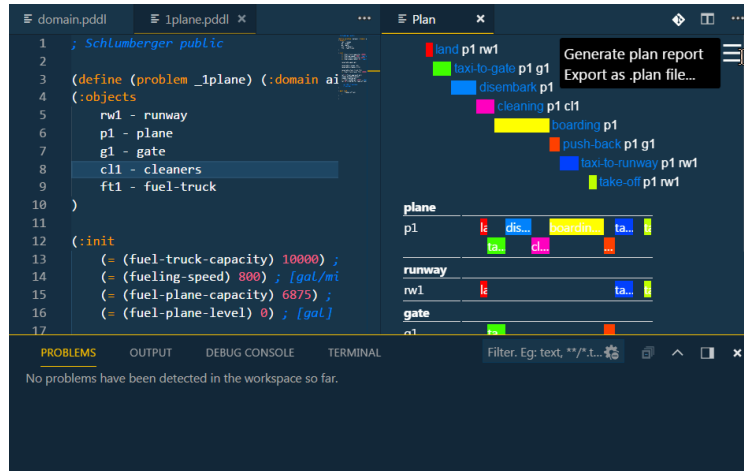


- Search against an adversarial.
- Zero-sum games.
- Basic algorithm: **Minimax**
- Extended techniques:
 - DFS minimax.
 - Alpha-beta pruning.
 - H-Minimax (evaluation functions)
 - Expectiminimax.

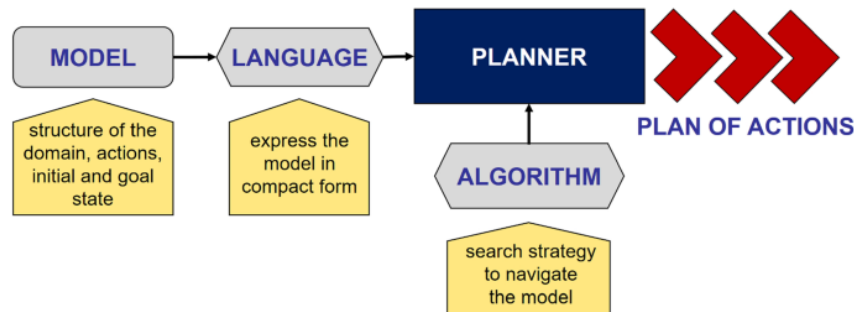


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Planning to act



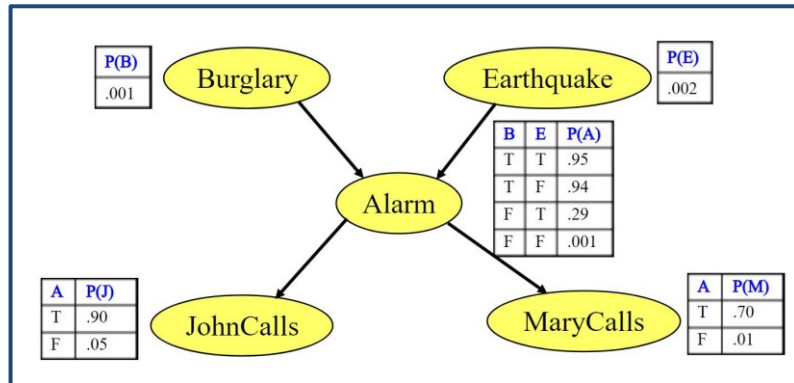
- How to build plan of actions for intelligent systems.
 - Sequential decision making.
- Combines search with KR
- KR for modeling the domain:
 - Domain features/variables.
 - Actions: precondition + effects
- PDDL & STRIPS as modeling languages.
- Many planners: FF, Fastdownard, FOND-ASP, etc.
- Extract domain-independent heuristics.
 - e.g., GraphPlan



Probabilities



- Beyond boolean values.
- Model and estimate chances of events.
- Probability as special functions.
 - Axioms of probabilities.
- Topics:
 - Joint probabilities.
 - Conditional probabilities.
 - Inference by enumeration.
 - Marginalization.
 - Bayes rule.
- Bayesian Networks: deals with complexity
 - Modeling and inference technique
 - Exploits domain independences.



Side effects of taking AI...

- Learn Python or get better at it.
- Learn better development practices (including git!)
- Learn about LaTeX.
- Improve teamwork skills.
- Make friends and future colleagues!
- *<your favourite side-effect>*



The Questions

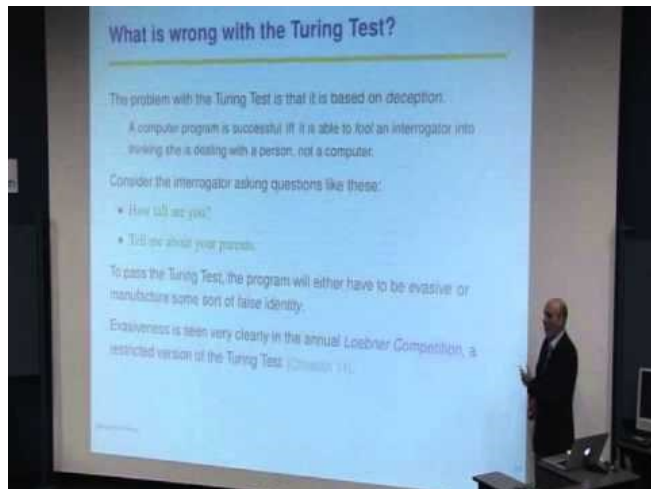


What would you tell a fellow student who is interested in the course?

Do you think you will be a better Computer Scientist after taking AI?



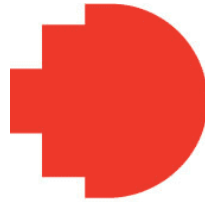
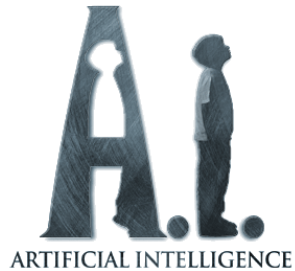
Two last videos to discuss...



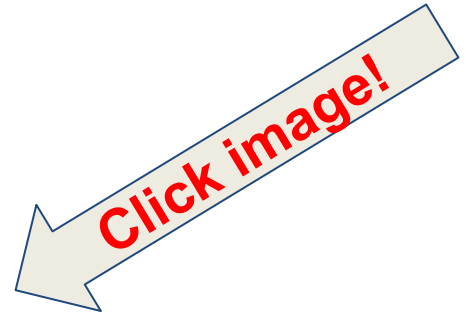
Prof. Hector Levesque
On the Turing Test



Prof. Moshe Vardi
Ethical Crisis in CS?



RMIT





Slides are adopted from Prof. Sebastian Sardiña